

Five Decades of Great Lakes Ice Cover: Sub-lake Scale Trends and Relationships with Meteorological Data

James Kessler¹ David Cannon² Alisa Young¹ Jia Wang¹

¹NOAA Great Lakes Environmental Research Lab ²Cooperative Institute for Great Lakes Research

Introduction

Ice cover in the Great Lakes has significant influence on regional climate, ecology, and economy and has been recorded with high accuracy since the advent of remote-sensing data in the early 1970's. This data is spatially resolved at 1.25KM resolution and available as daily files throughout each ice season¹.

Previous work has assessed large-scale (i.e. lake or basin-wide) ice cover fluctuations through time and their relationship to global weather phenomena^{2,3,4,5}. Mason et al. (2016)⁶ used the Great Lakes Aquatic Habitat Framework (GLAHF)⁷ to analyze ice cover duration for 1973-2013. The delineation of subregions in GLAHF includes multiple spatial scales and was designed by considering physical, ecological and socioeconomic factors.

Since 2013, the Great Lakes have seen two extreme highs (2014 and 2015) and two extreme lows (2024 and 2017) in the top 5 highest/lowest for seasonal max ice (see fig. 1). This work provides an update to Mason et al. 2016 for recent years, analysis of two additional ice metrics (seasonal max and mean), and correlations to several meteorological variables from 530 met stations.

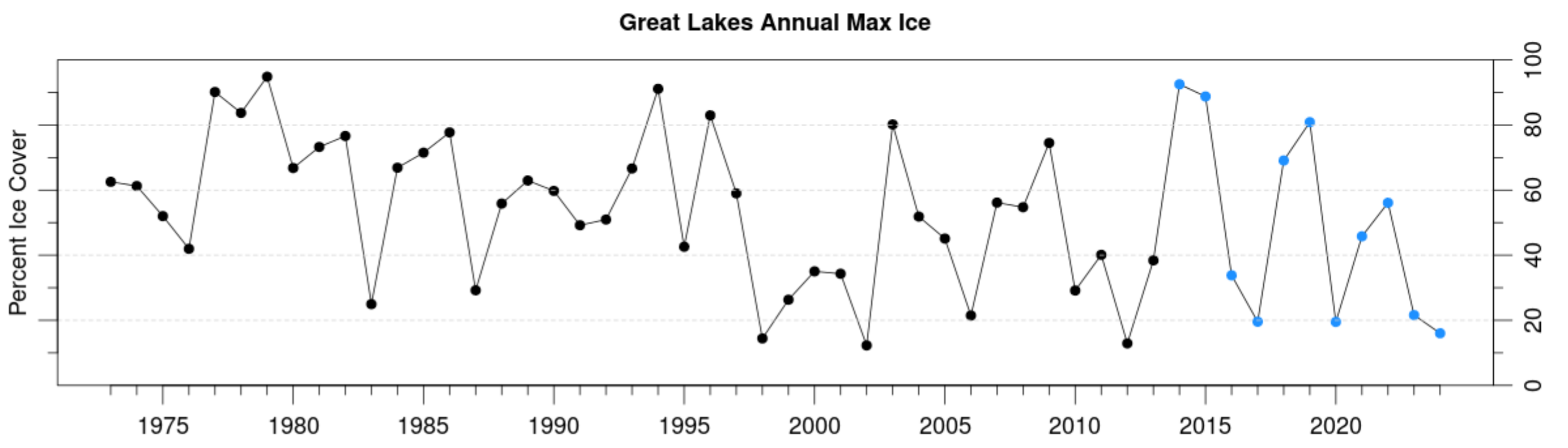


Figure 1. Annual Max Ice cover for the Great Lakes (ice years after 2013 highlighted in blue).

Methods

Research Questions:

1. Are there statistically significant ice trends at these spatial scales?
2. How do ice cover metrics correlate to nearby meteorological station data?

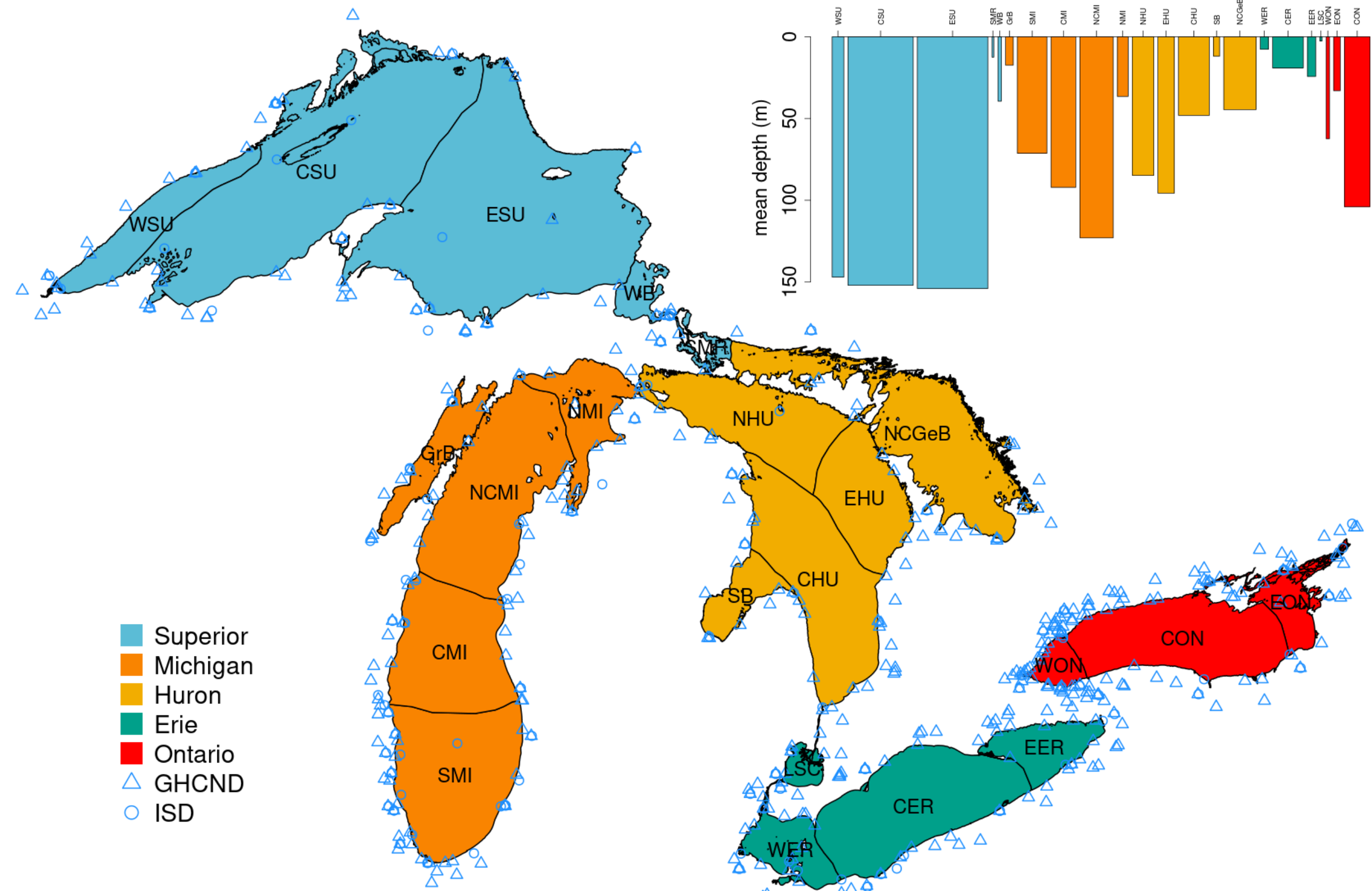


Figure 2. Spatial delineation of GLAHF subregions over water in relation to selected met station locations from the Global Historical Climatology Network Database⁸ and the Integrated Surface Database⁹. The upper right figure shows depth and (relative) surface area of each subregion in the y and x directions, respectively.

To address the first research question, the below metrics are calculated for each GLAHF subregion and ice season and then Mann-Kendall monotonic trend test¹⁰ is applied to check for statistical significance:

- **Annual Max Ice Cover (AMIC):** maximum percent (single daily value) during a given season
- **Seasonal Average Ice Cover (JFM):** mean daily percent of ice cover for Jan 1 - Mar 31
- **Ice Duration:** Total (non-consecutive) days[†] where ice cover exceeded 10%

To address the second research question, station data^{8,9} is aggregated for each subregion based on a 20 KM radius and correlations between ice metrics and daily mean meteorological surface variables are identified. On average, 30 stations per subregion were located within the radius but station data availability varied through time, location and variable. Temperature min/max were reported most frequently (median station count $\approx$ 10 per subregion). Other variables—cloud cover, precipitation, wind speed, were reported less frequently (median $\approx$ 5).

†: days are only counted Jan 1 - Mar 31 due to shorter ice reporting windows during 1970's and 80's. This method likely *underestimates* trends in decreasing ice duration but is required to avoid artificial identification of season length decrease.

Analysis

Part 1: Trend Analysis

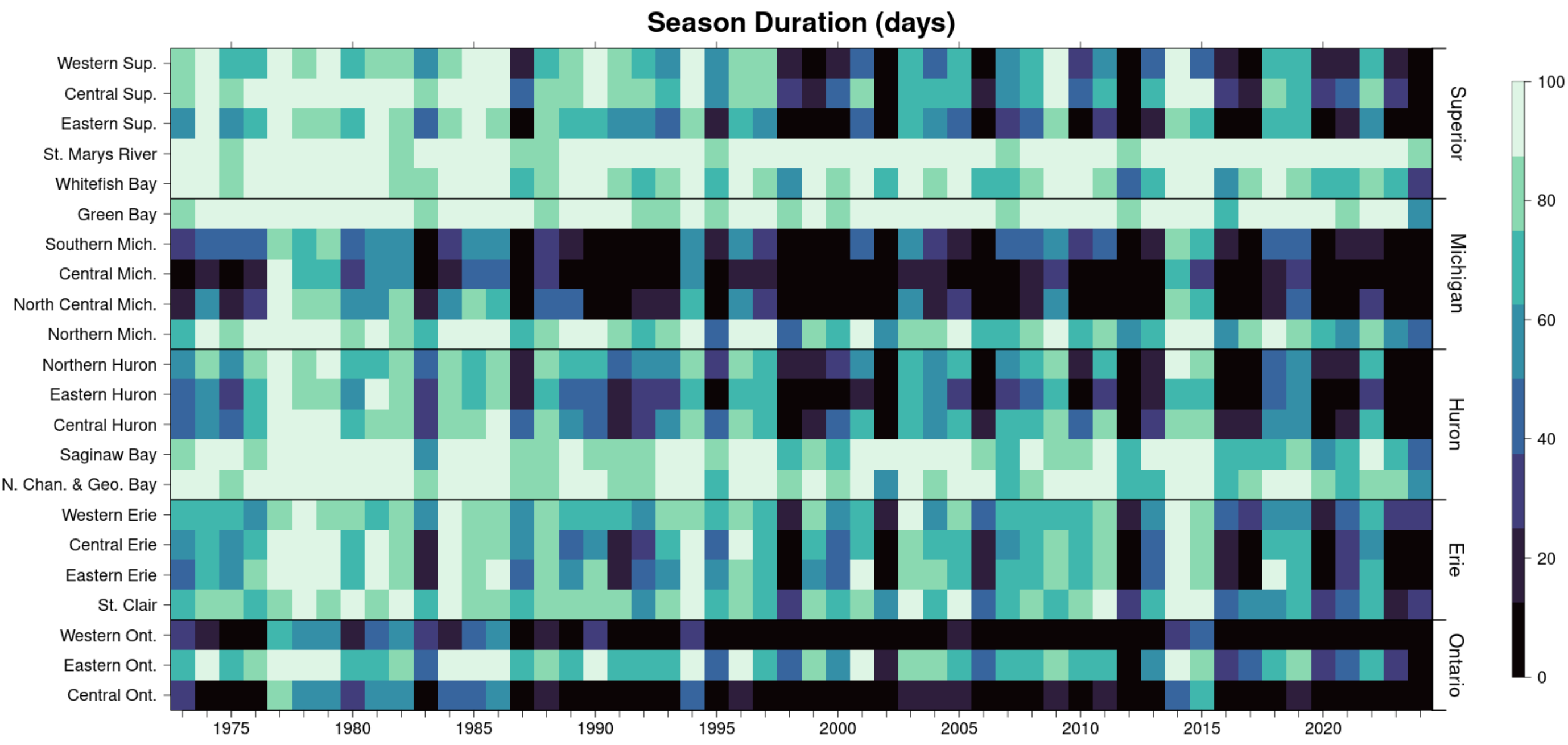


Figure 3. Ice duration for each subregion.

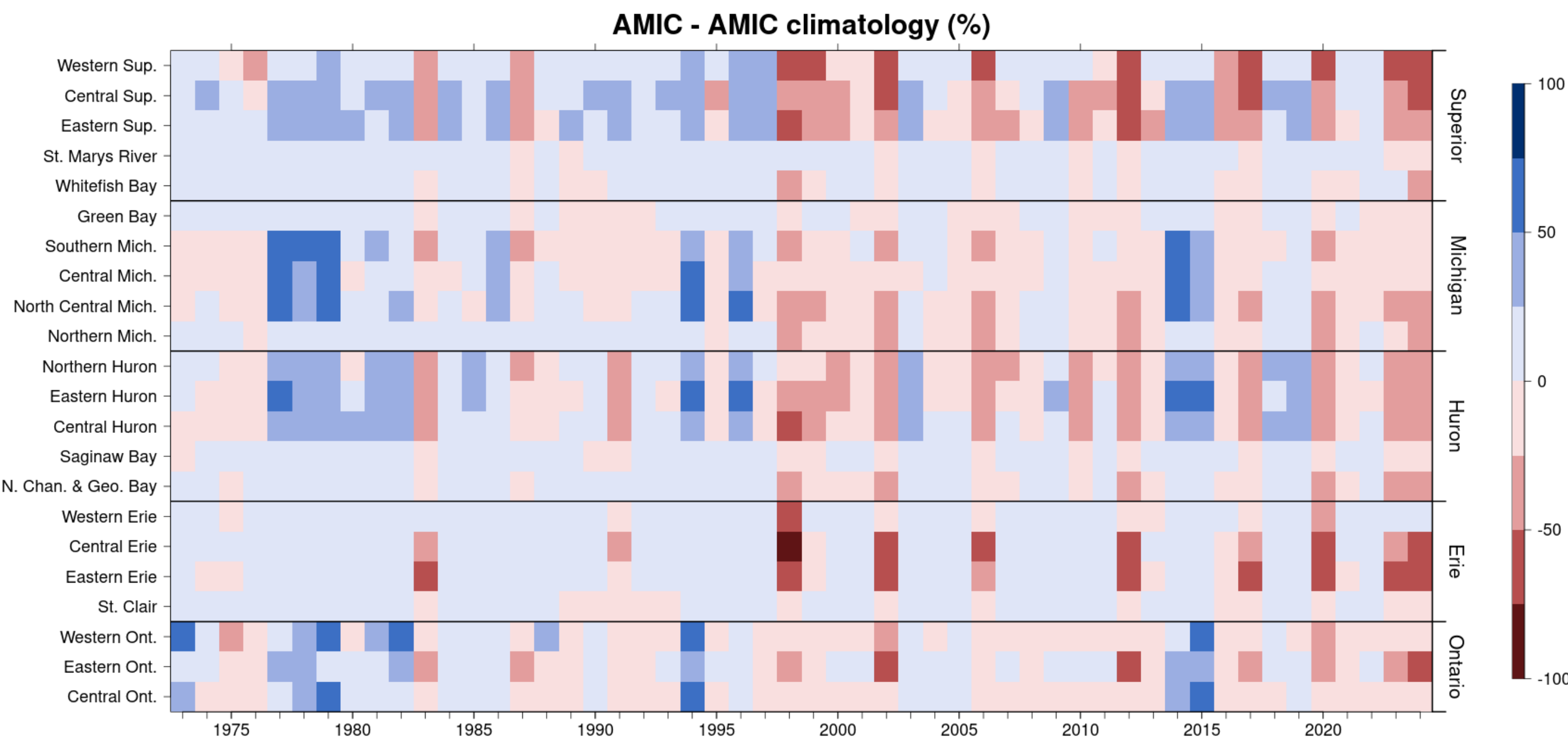


Figure 4. Departure from climatology for annual max ice cover for each subregion.

Season duration decrease is visible in some regions but not apparent in others (see fig. 3). Regions that show decrease are often offshore and deep: Superior (WSU, CSU, ESU), Huron (NHU, EHU, CHU), and Ontario (WON, CON). This may be due to warmer subsurface temps^{11,12} during later years providing heat to surface waters and preventing freezing, whereas shallow regions may be able to consistently release all heat from the water column. (Despite the lack of obvious decrease, a majority of regions do show significant decreasing trends in fig. 5).

Annual max ice departures (fig. 4) show similar features to ice duration—shallow, isolated regions (e.g. GrB, SB, WB, SMR) show little variation through time. The 1997-98 changepoint identified by Mason et al.⁶ is prominent in both figures above (transition from white to black and blue to red).

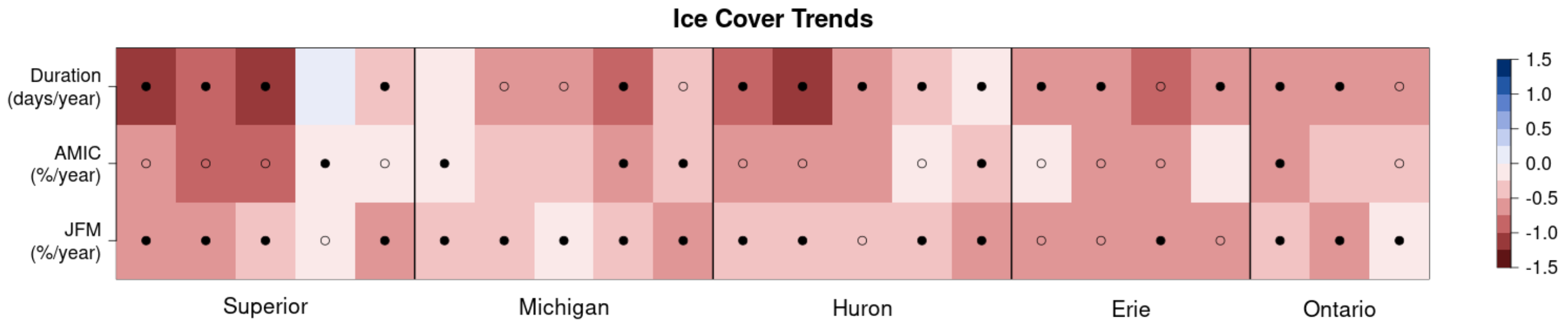


Figure 5. Linear trends for each metric. Open (solid) dots indicate 95% (99%) statistical significance. Columns correspond to rows in figures 3 and 4.

Part 2: Correlation with Met Stations

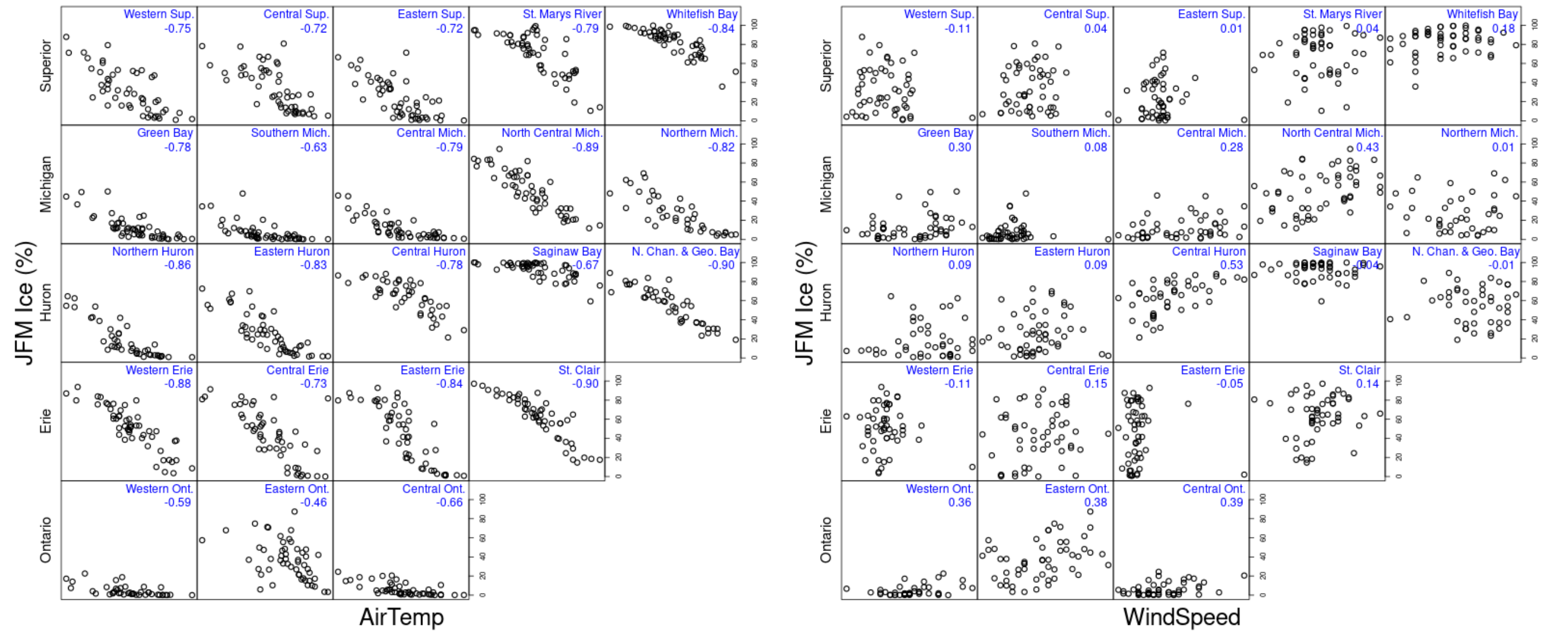


Figure 6. Correlation between Seasonal Ice Cover and Air Temperature (left) and Wind Speed (right). Each point denotes a season where both ice and meteorologic variable was averaged over JFM from daily values.

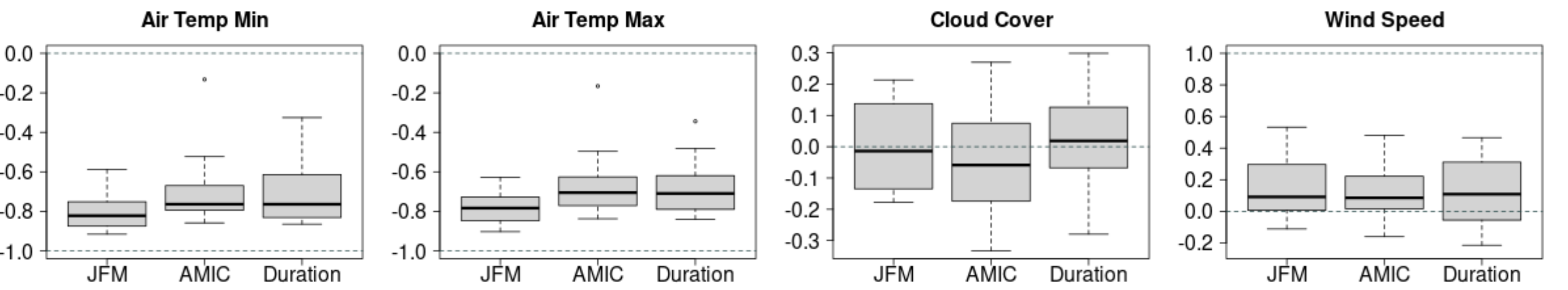


Figure 7. Distribution (N=22 subregions) of correlations (Pearson's r) between three ice metrics and met station variables.

As expected, ice cover showed strong inverse correlation with air temperature. Wind speed showed moderate correlation to ice in some regions which may be due to wind's influence on heat distribution in the water column via mixing (though this impact is more relevant during fall overturn). The analyses presented here are a small fraction of possible correlation pairs and more work is needed, including different time periods and other metrics like ice onset/offset.

Summary

Trend assessment:

- A majority of subregions show significant decreasing trends in all three metrics
- Seasonal average trends were significant in ALL 22 subregions
- Max Ice Cover showed the least significant trends (17/22 regions)

Met Data Correlation:

- Air temperature was strongly anti-correlated with ice cover for all metrics
- More work needed: comprehensive assessment of ice cover and met data correlations

References

[1] Ting-Yi Yang et al. "A consistent Great Lakes ice cover digital data set for winters 1973–2019". In: *Scientific Data* 7.1 (Aug. 2020). DOI: 10.1038/s41597-020-00603-1.

[2] Sapna Sharma et al. "Widespread loss of lake ice around the Northern Hemisphere in a warming world". In: *Nature Climate Change* 9.3 (Jan. 2019), pp. 227–231. DOI: 10.1038/s41558-019-0393-5.

[3] Catherine M. O'Reilly et al. "Rapid and highly variable warming of lake surface waters around the globe". In: *Geophysical Research Letters* 42.24 (Dec. 2015). DOI: 10.1002/2015gl066235.

[4] Jia Wang et al. "Decadal Variability of Great Lakes Ice Cover in Response to AMO and PDO, 1963–2017". In: *Journal of Climate* 31.18 (Sept. 2018), pp. 7249–7268. DOI: 10.1175/jcli-d-17-02883.1.

[5] Xuezhai Bai et al. "Interannual variability of Great Lakes ice cover and its relationship to NAO and ENSO". In: *Journal of Geophysical Research: Oceans* 117.C3 (Mar. 2012). DOI: 10.1029/2010JC006932.

[6] Lacey A. Mason et al. "Fine-scale spatial variation in ice cover and surface temperature trends across the surface of the Laurentian Great Lakes". In: *Climatic Change* 138.1–2 (June 2016), pp. 71–83. DOI: 10.1007/s10584-016-1721-2.

[7] Lizhu Wang et al. "A spatial classification and database for management, research, and policy making: The Great Lakes aquatic habitat framework". In: *Journal of Great Lakes Research* 41.2 (June 2015), pp. 584–596. DOI: 10.1016/j.jglr.2015.03.017.

[8] Matthew J. Menne et al. "An Overview of the Global Historical Climatology Network-Daily Database". In: *Journal of Atmospheric and Oceanic Technology* 29.7 (July 2012), pp. 897–910. DOI: 10.1175/jtech-d-11-00103.1.

[9] Adam Smith, Neal Lott, and Russ Vose. "The Integrated Surface Database: Recent Developments and Partnerships". In: *Bulletin of the American Meteorological Society* 92.6 (June 2011), pp. 704–708. DOI: 10.1175/2011bams3015.1.

[10] Henry B. Mann. "Nonparametric tests against trend". In: *Econometrica: Journal of the econometric society* (1945), pp. 245–259.

[11] David Cannon et al. "Investigating Multidecadal Trends in Ice Cover and Subsurface Temperatures in the Laurentian Great Lakes Using a Coupled Hydrodynamic–Ice Model". In: *Journal of Climate* 37.4 (Feb. 2024), pp. 1249–1276. DOI: 10.1175/jcli-d-23-0092.1.

[12] Eric J. Anderson et al. "Seasonal overturn and stratification changes drive deep-water warming in one of Earth's largest lakes". In: *Nature Communications* 12.1 (Mar. 2021). DOI: 10.1038/s41467-021-21971-1.